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Optimizing Smart City Water Distribution Systems using Deep Reinforcement Learning

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Abstract—Inefficient scheduling in water distribution systems can lead to energy waste, costly overflows, and a system that cannot keep up with demand. Simultaneous real-time management of system components such as pumps and valves to optimize operation in response to demand variations is a challenging task. Recent advances in deep reinforcement learning provides an opportunity to overcome the state explosion problem using function approximation to generalize from a limited interaction with the environment. In this work, we train a Long Short-Term Memory (LSTM) based Reinforcement Learning (RL) agent to optimize the energy usage of a smart water distribution system while maintaining a safe operating envelope. We compare the performance of the RL agent to two agents based on human experience in the domain; a baseline controller that is based on simple operational logic, and a fuzzy logic controller that captures imprecise human requirements. We show that the RL agent outperforms the other agents in terms of energy usage and operational safety, indicating its potential benefits for large-scale smart city systems. Future research work will focus on prioritized large-scale system scheduling to cope with smart city emergency situations.

Index Terms—Water distribution, Smart city, Reinforcement learning, Optimization, Fuzzy logic control, Scheduling.

I. INTRODUCTION

Every day, the United States uses 322 billion gallons of water, of which 87% is freshwater. Unfortunately, increasing demand has increased power consumption, 2% of overall electricity usage, a 52% increase since 1996 [1]. Over 80% of municipal water processing and distribution expenditures are attributable to energy charges, with an average of 3,300 to 3,600 kWh needed to treat and transport one million gallons of water [2]. The financial and ecological toll is high for these types of operations. Water conservation, efficient technology adoption, and public awareness campaigns are all essential parts of the solution to this problem [3]. Implementing these measures can reduce water management's energy footprint while building a more sustainable future.

As clean water is essential to human health and progress, the government should emphasize its distribution and management with higher importance. However, it might be challenging to keep up with the needs of a rapidly expanding metropolitan population while providing an efficient service at a reasonable price [4]. Pipelines, reservoirs, treatment plants, and pumping stations have varying degrees of efficiency based on different

factors, including distance from populated areas, the amount of energy needed to operate, and the overall cost. The demand for reliable data to avert ecological and monetary losses due to climate change-related water shortages is increasing [5]. To address these challenges, employing smart technologies, revitalizing aging infrastructure, and developing cross-domain applications are imperative [6].

Real-time scheduling of a water distribution system requires intricate and time-consuming computations. To keep tank levels within upper and lower bound constraints while fulfilling user demands and system requirements, however, requires precision pump management, which has the potential to save operational energy. Optimizing water distribution systems concerning pumps and valves for greater efficiency requires considering the already-present circumstances of demand fulfillment and reservoir-level management, which can lead to significant cost savings [7].

Pump and valve scheduling can optimize water distribution system operations by determining control rules or time-based specifications to fulfill supply or demand requirements at the lowest feasible energy cost [8]. However, water distribution systems in large cities pose challenges that need more sophisticated solutions [9]. Pump scheduling along with valve configuration optimization is one example of a problem that calls for such solutions [10].

The pump scheduling problem aims to maximize water supply efficiency while satisfying all safety requirements and reducing energy usage. The particular approaches used vary depending on the specifics of the water distribution network. One way to reduce operational costs is to use water storage tanks and run pumps during off-peak electrical hours [11]. Pumps should only be turned on and off sparingly, as doing so too frequently might wear out the machinery and lower the water quality in the storage tanks [12].

The primary goal of this study is to assess the efficacy of the deep reinforcement learning (DRL) approach for scheduling pumps and valves in smart water distribution systems while satisfying all restrictions. These methods include estimating customer demand, employing real-time datasets, utilizing expert knowledge, comparing with baseline controllers, and assessing them in different simulations. In addition, we also compared this DRL approach with fuzzy logic and baseline

controllers to evaluate its performance comprehensively.

The main contributions of this paper are:

- 1) Building and Training a reinforcement learning agent for a smart water distribution system while considering multiple factors, such as pump cost in the reward function, including the number of pumps and valves used for the cumulative reward.
- 2) Considering equipment control at both ends of the distribution network (smart pumps and isolation valves) to emulate real-time situations for our ongoing testbed [13].
- 3) Evaluating the suggested method against the standard controller methods about the total cumulative amount earned.

This paper is organized as follows: Section II provides a detailed literature assessment of related studies of optimizing smart water distribution networks. The problem of pump scheduling of a water distribution system is formulated in Section III. Section IV outlines the simulation results and discusses the experimental outcomes. The work is concluded in Section V.

II. LITERATURE WORKS

Historically, steady-state optimization methods have been used to reduce energy consumption or improve system performance when dealing with water pressure in water distribution networks (WDNs). The authors in [14]–[16] have optimized several decision factors, including pump flows, pressures, speeds, and tank water levels. Steady-state models are less flexible since they cannot accommodate the unpredictable nature of water demand. Including real-time data such as flow rates, pressures, and tank levels could lead to more accurate models and enable better pump scheduling and optimization based on actual conditions, overcoming these restrictions [17].

Pump scheduling optimization in water distribution networks is a growing field, and model predictive control (MPC) has emerged as a viable solution [18]. The steady-state water flow problem and optimal service at the junction node (Junctions are water-consuming nodes) were addressed in [18], where authors suggested a fixed-point iteration technique based on a neural network. Using techniques from graph theory, mixed-integer programming, and convex relaxations, researchers in [19] devised a mixed-integer quadratically restricted quadratic program solution to the water flow problem. To deal with the unknowns inherent in WDNs, authors in [20] suggest a stochastic MPC technique, which is applied to the water supply system by utilizing precise information on consumer demand. Similarly, [21] suggests a nonlinear economic MPC technique that uses a two-layer control system with hourly control intervals and a pump scheduling approach with a one-minute sampling period.

The current research revealed several approaches to optimizing a water distribution system [22]. In addition, recent developments in deep learning and reinforcement learning have been applied to the issue of pressure control in WDNs. Results from using deep neural networks to predict future water needs have been encouraging [23]. For example, to

predict the 15 min-future demand, the authors in [24] suggest an approach based on a gated recurrent unit network that can predict up to 24 hours into the future. Meanwhile, deep reinforcement learning has recently been investigated as a potential solution to the pump scheduling problem, which is nonlinear and nonconvex. The optimization issue was described as a Markov decision process in [7], and the authors suggested a multiobjective deep reinforcement learning-based solution for the real-time optimization of pumps. In addition, the authors employed a genetic algorithm to improve work scheduling and reduce the amount of polluted water drank [25].

Due to its multiple benefits, reinforcement learning (RL) has become a popular method for resolving scheduling challenges in various fields. Researchers created a two-stage RL-based method to address the challenge of scheduling data-intensive tasks on a unified data grid [26]. RL's versatility and adaptability make it a viable approach to scheduling challenges [27]. The authors concentrated on minimizing energy expenditures while satisfying operational restrictions, including fulfilling demand, maintaining pressure to prevent leaks and burst pipes, and maintaining reservoir levels to avoid overflow [28].

In this paper, we have developed and tested a reinforcement learning agent for a water distribution system. We considered a wide range of variables, such as pump cost in the reward function, including the number of pumps and valves utilized and the current price of water. This is essential since it assesses various criteria while training an RL agent for a water distribution network. We have also considered end-and beginning-of-line treatment equipment, including pumps and isolation valves, to more accurately model practical situations. Then, the agent has been trained to efficiently optimize water distribution in light of the cost of water at any given time. In conclusion, we have compared DQN approach to some baseline and fuzzy logic controller approaches and discovered that our proposed strategy is more effective than those heuristic domain methods for cumulative reward.

III. PROBLEM FORMULATION

The water pump's practical and dependable operation is essential for a steady water supply. A pump scheduler makes strategic, long-term control decisions that boost system efficiency. However, two physical aspects of the system need to be fixed. The first is that adding more pumps reduces pumping efficiency. Adding a second pump decreases (not two times of a single pump) flow significantly while increasing power consumption by about 100%. Second, while running pumps at low tank levels can improve energy efficiency, it also raises the danger of emptying the tank during abrupt surges in demand, reducing water pressure for system users. An efficient pump scheduler can make control decisions that optimize energy savings while reducing the risk of exceeding critical limits [29]. The fluid pace that moves through the system depends on the number of active pumps and the volume of fluid currently stored in the tank. The flow rate for a single pump and an empty tank is much greater than that for three pumps and the same empty tank. At near-maximum tank

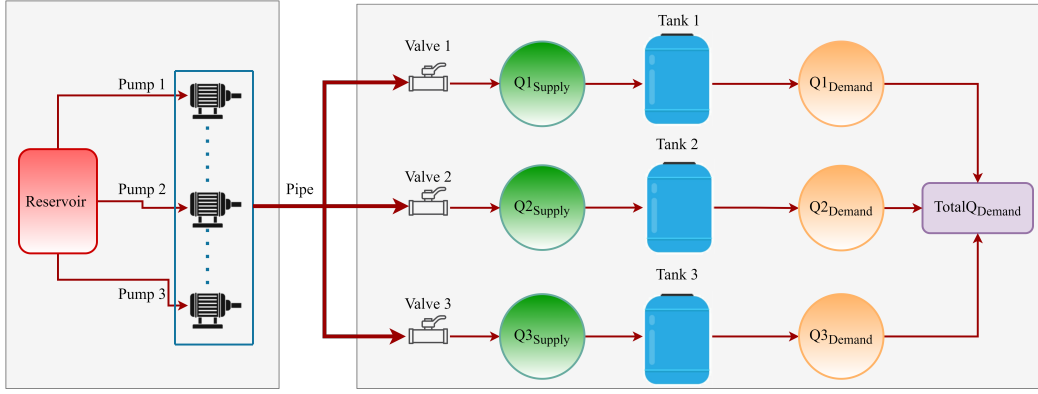


Fig. 1. Proposed real-time water Scheduling model based on Reinforcement Learning

capacity, the interaction between the pump and system curves causes the system curve to behave non-linearly [29].

A. System Description

We have created a reinforcement learning (RL) method to find the best way to schedule pumps in a water distribution system, enhancing efficiency and economy. The RL agent may dynamically modify the number of pumps and valves running to optimize energy consumption while fulfilling use needs by considering energy usage, demand requirements, and the influence on costs [29], [30]. Fig. 1 provides the architecture of the smart water distribution system, where we have one reservoir, three pumps, three valves, and three tanks. We have considered three tanks dedicated to three buildings. Optimizing the valves and pumps based on the water level in these tanks using reinforcement learning agents and comparing them with two different approaches will be this study's primary idea of investigation.

different for all three tanks. Equation 1 represents the change in water level (h) over time (t), which is directly influenced by the difference between the inflow and outflow rates.

$$A \cdot \frac{dh}{dt} = Q_{Supply}(t) - Q_{Demand}(t) \quad (1)$$

We employ a time-varying mean and random white noise to simulate the changing nature of water demands and supplies. Water demand at time t , $Q_{Demand}(t)$, is computed explicitly as the sum of a time-varying mean, $\mu(t)$, plus a random white noise component, $\eta(t)$. We assume the random noise is typically distributed over the interval $[-25, 25]$. This introduces randomness into the water demand model by allowing the noise component to take any value in the interval $[-25, 25]$ with equal probability. The water demand over a 24 hour period is a function of time given as by the Equation 2, where $\mu(t)$ is the expected demand, and $\eta(t)$ represents the demand uncertainty, which is sampled from a uniform distribution where $a = -25$ and $b = 25$.

$$Q_{Demand}(t) = \mu(t) + \eta(t) \quad (2)$$

We utilized a simple basic controller model and fuzzy logic model controller to predict flow rates in the water distribution system for a given combination of pumps and valves rather than building a complicated static or dynamic model. It helped us rapidly test new valve and pump configurations and teach our RL agent about new challenges. The supply is determined by the number of pumps running, $a \in 0, 1, 2, 3$ and number of valves open at that specific time, $v \in 0, 1, 2, 3$ based on the water system curve [29]. The relationship among the flow rate, number of pumps, a , and number of valves, v , is provided in Table I.

Power consumption is defined as the number of pumps running, a , and c is the cost of running the pump per hour for simplifying the problem. Equation 3 is the reward for this environment. An additional cost is added if the water height is close to the maximum or minimum water levels, h_{max} or h_{min} , respectively, to avoid overflowing or emptying the tank [29], [30].

$$r(h, a, c) = -10 \times \mathbb{1}_{h \geq (break1)}(h) - 5 \times \mathbb{1}_{(break2)}(h) - a \cdot c \quad (3)$$

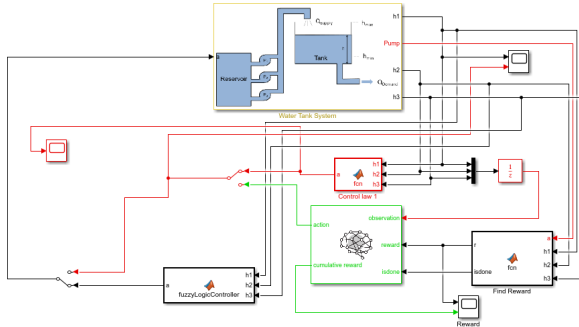


Fig. 2. Water Distribution System Simulink model

Fig. 2 shows the Simulink model of the Water Distribution System, and Fig. 2 illustrates the relationship between water demand and water height for three tanks (h_1 , h_2 , and h_3 are the water height for three tanks, respectively). Equation 1 describes the flow rate of water in the system. Where $A = 10m^2$ and maximum height is 7m. $\frac{dh}{dt}$ is the flow rate and $Q_{Supply}(t)$ is the water flow rate, which is mapped with number of pumps and valves for three tanks. $Q_{Demand}(t)$ is

TABLE I
SUPPLY FUNCTION MAPPING WITH RESPECT TO A AND V

$\begin{array}{c c} & v \\ \hline a & \end{array}$	0	1	2	3
0	0	0	0	0
1	0	164	82	54.67
2	0	246	123	82
3	0	410	205	136.67

where $\mathbb{1}$ is the indicator function. The cost associated with energy consumption is c and we used a constant value for this experiment. We have used a customized function to input the simulink model to generate a water demand profile based on the number of days considered. Fig. 3 provides the graphical representation of the daily demand curve of the dataset.

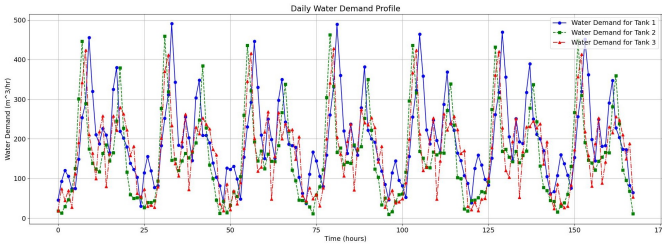


Fig. 3. Dataset Description

B. Deep Reinforcement Learning Agent Architecture

The objective of the reinforcement learning agent is to schedule the number of pumps and valves running to minimize the system's energy usage and satisfy the water height requirements for three tanks [29], [30]. We have developed a Deep Q-network (DQN) agent that leverages a critical Q-value function to approximate long-term rewards. Specifically, we created a deep neural network for the critic and used a recurrent neural network by setting the useLSTM parameter to true. We then specified various options for the agent, such as a sequence length of 100, which should help inform the agent's decision-making process. The action information consists of 8 values, written as $[0,0,1,1]$. The first number indicates the number of active pumps, while the second, third, and fourth values represent the on/off states of the respective valves. We propose using binary or category variables rather than a single vector to capture the action information. As a result, the training process's efficiency may be improved by decreasing the input dimensionality. The experimental observations we utilize to train our RL agent comprise the heights of the water in three tanks at various times. We use a Long Short-Term Memory model to train the agent. This model can capture temporal relationships in sequential data, so it is well suited to our purpose. Three fully linked layers and two Rectified Linear Unit (ReLU) activation layers make up our LSTM model. We combine action and observation data to train our RL agent and run it through an LSTM model with exemplary architecture.

IV. SIMULATION RESULTS

A. Baseline controller

We have implemented a primary controller for our water distribution system that uses simple calculations to determine the optimal number of valves to open and the appropriate pump settings. The four rules described below specify the simple control mechanisms of a water distribution system based on the readings of three water level heights (h_1 , h_2 , and h_3) and specific threshold values (break1 and break2).

- All three tank heights are within the specified range (between break2 and break1), and the output of the rule is $[0,0,0,0]$, which means that no pump is running, and no valves are open.
- Two tanks (h_1 and h_2) are within the normal range, but h_3 is not (below break2 or above break1), then the system is in a partial failure state. The rule's output is $[1,0,0,1]$, which means that one pump is running, and only the third valve is open.
- One tank (h_1) is within the normal range, but h_2 and h_3 are not (below break2 or above break1), then the system is in a partial failure state. The rule's output is $[2,0,1,1]$, which means two pumps are running, and the second and third valves are open.
- All three tank heights are outside the normal range (below break2 or above break1), then the system is in a complete failure state, and the output of the rule is $[3,1,1,1]$, which means that three pumps are running, and all three valves are open.

B. Fuzzy Logic controller

Based on water height levels in three tanks (h_1 , h_2 , and h_3), a fuzzy logic controller makes decisions (pump on or off) concerning running pumps and valves. Considering the varied water levels in the tanks, the objective is to handle the water schedule efficiently. The controller first defines membership functions for each height (h_1 , h_2 , and h_3). Each variable has its own "Low," "Middle," and "High" fuzzy set. We then fuzzified the membership functions for each height. The water levels are measured to calculate membership values for the linguistic categories ('Low,' 'Mid,' and 'High'). The generated fuzzy values reflect how well each variable fits into these conceptual frameworks regarding language use. Decisions about the functioning of pumps and valves are made using the fuzzified input variables in conjunction with the controller's inference rules. The rules evaluate combinations of these fuzzy values to determine the proper controls. For instance, if both h_1 and h_2 are 'Low' or 'Mid,' and h_3 is 'Low' or 'Mid,' this may mean that all storage tanks are full, all pumps are running, and all valves are open. In contrast, if both h_1 and h_2 are 'Low' or 'Mid,' but h_3 is 'High,' this might imply that tank 3 is overflowing, necessitating just two pumps and the closure of valves 3. These inference rules provide a structured way to decide based on the water level conditions. From the results of the inference rules, the controller derives the control values, such as the number of pumps to operate

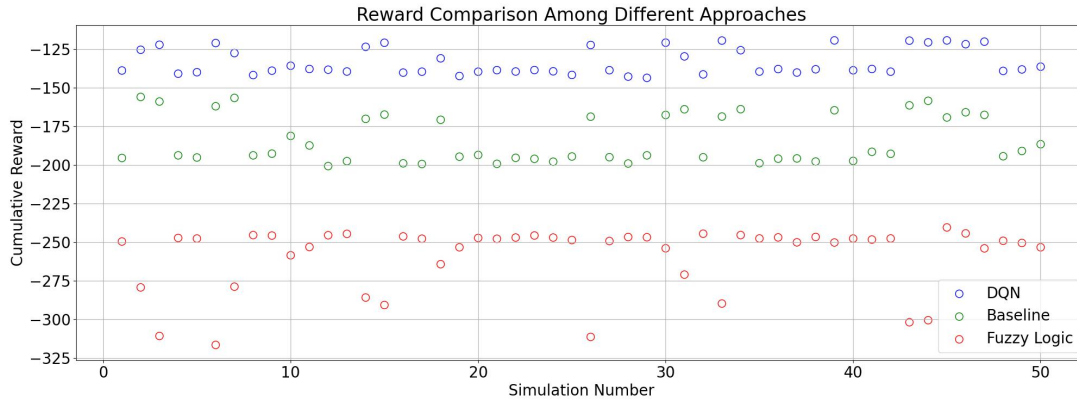


Fig. 4. Cumulative rewards for both the agent and the baseline controller

and each valve's open/closed status. In this fuzzy logic system, the fuzzify function calculates a value's membership level in a triangle membership function. Fuzzy logic controllers can perform differently from the baseline controller for various reasons, including insufficient tuning, flawed input variables, inadequately designed rules, and extreme problem complexity.

C. Reinforcement Learning controller

To train the agent, we set training options such as running training for 100 episodes, each lasting 168 time steps, and displaying the training progress in an Episode Manager dialog box. We use the train function to train the agent, which can be computationally intensive and may take several hours to complete. However, we can also load a pre-trained agent by setting doTraining to false to save time. It is essential to tune the agent's parameters to ensure optimal performance [30]. Fig. 2 describes the Simulink model of the water distribution system we have used in this experiment.

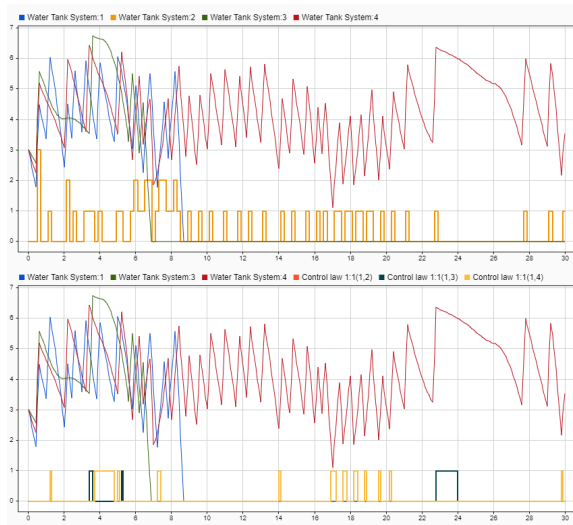


Fig. 5. Explanation of Control Actions

To validate the performance of the trained agent, we conducted simulations in a water-tank environment using the rlsimulationOptions and sim functions. We connected the RL Agent block to simulate the agent's performance by toggling the manual switch block. We set the maximum number of steps for each simulation and the number of simulations. We repeated this experiment for 50 simulations. The environment reset function was set to initialize a different water height and demand in each simulation. To ensure a fair comparison with the baseline and fuzzy logic controller, we reset the initial random seed used in the environment reset function to ensure that both the agent and the controller operate under the same conditions [30]. In addition, we initialized cumulative reward result vectors for both the agent, baseline, and fuzzy logic controller and plotted the cumulative rewards to compare the performance of both approaches visually [30].

In Fig. 4, it is apparent that the DQN agent invariably exceeds both the baseline and fuzzy logic controllers in all simulations, mainly based on the cumulative reward. Each tank's water level (h_1 , h_2 , and h_3) and the number of running pumps (a) are shown by the yellow, green, red, and blue lines, respectively, in Fig. 5. In the first subplot, we see how the control actions vary as a function of the water levels in the tanks, and in the second subplot, we see how the water levels in each tank change when the valves are opened and closed.

V. CONCLUSIONS & FUTURE WORKS

This paper provides a method for optimum pump and valve scheduling in water distribution systems by utilizing Long Short-Term Memory (LSTM) based Reinforcement Learning (RL) to learn the appropriate scheduling strategy from interactions with the environment by training the agents. We demonstrate that the proposed method is superior by contrasting the outcomes of employing baseline and fuzzy logic controller approaches in water distribution systems. Applying RL to water distribution systems presents a promising approach to address wasteful energy use, high costs, and inadequate system performance, leading to increased productivity and improved sustainability. To effectively manage a system, it is vital to

be aware of both the benefits and drawbacks of optimization. Nonetheless, the encouraging findings of this work indicate that LSTM-based RL is a realistic and successful technique for improving water distribution system scheduling. Future work directions for optimizing the scheduling of a water distribution system scheduling include the following:

- Incorporating differentiated water demand to reflect variations in demand across different tanks,
- Exploring the use of multiple RL agents to optimize various aspects of the system,
- Improving the model architecture by incorporating attention mechanisms or advanced optimization techniques to capture temporal dependencies and make better scheduling decisions,
- Developing a real-time implementation of the LSTM-based RL model for a more responsive and effective water distribution system.
- Prioritizing scheduling for large-scale smart water distribution systems.

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REFERENCES

- [1] S. Pabi, A. Amarnath, R. Goldstein, and L. Reekie, "Electricity use and management in the municipal water supply and wastewater industries," *Electric Power Research Institute, Palo Alto*, vol. 194, 2013.
- [2] C. Copeland and N. T. Carter, "Energy-water nexus: The water sector's energy use," 2014.
- [3] Center for Sustainable Systems, "U.S. Water Supply and Distribution Factsheet." <https://css.umich.edu/publications/factsheets/water/us-water-supply-and-distribution-factsheet> (accessed March, 09, 2023).
- [4] G. Kallis and H. Coccossis, "Water for the city: lessons from tendencies and critical issues in five advanced metropolitan areas," *Built Environment (1978-)*, pp. 96–110, 2002.
- [5] G. Sun, S. G. McNulty, J. A. Moore Myers, and E. C. Cohen, "Impacts of multiple stresses on water demand and supply across the southeastern united states 1," *JAWRA Journal of the American Water Resources Association*, vol. 44, no. 6, pp. 1441–1457, 2008.
- [6] A. Predescu, C.-O. Truică, E.-S. Apostol, M. Mocanu, and C. Lupu, "An advanced learning-based multiple model control supervisor for pumping stations in a smart water distribution system," *Mathematics*, vol. 8, no. 6, p. 887, 2020.
- [7] G. Hajgató, G. Paál, and B. Gyires-Tóth, "Deep reinforcement learning for real-time optimization of pumps in water distribution systems," *Journal of Water Resources Planning and Management*, vol. 146, no. 11, p. 04020079, 2020.
- [8] M. Behandish and Z. Y. Wu, "Concurrent pump scheduling and storage level optimization using meta-models and evolutionary algorithms," *Procedia Engineering*, vol. 70, pp. 103–112, 2014.
- [9] N. Siddique and H. Adeli, *Computational intelligence: synergies of fuzzy logic, neural networks and evolutionary computing*. John Wiley & Sons, 2013.
- [10] S. M. Bunn and L. Reynolds, "The energy-efficiency benefits of pump-scheduling optimization for potable water supplies," *IBM Journal of Research and Development*, vol. 53, no. 3, pp. 5–1, 2009.
- [11] A. Candelieri, R. Perego, and F. Archetti, "Bayesian optimization of pump operations in water distribution systems," *Journal of Global Optimization*, vol. 71, pp. 213–235, 2018.
- [12] H. Donâncio, L. Vercouter, and H. Roelowski, "The pump scheduling problem: A real-world scenario for reinforcement learning," *arXiv preprint arXiv:2210.11111*, 2022.
- [13] N. Zohrabi, P. J. Martin, M. Kuzlu, L. Linkous, R. Eini, A. Morrisett, M. Zaman, A. Tantawy, O. Gueler, M. A. Islam, N. Puryear, H. Kalkan, J. Lundquist, E. Karincic, and S. Abdelwahed, "Opencity: An open architecture testbed for smart cities," in *2021 IEEE International Smart Cities Conference (ISC2)*, pp. 1–7, 2021.
- [14] J. Bene, I. Selek, and C. Hős, "Comparison of deterministic and heuristic optimization solvers for water network scheduling problems," *Water Science and Technology: Water Supply*, vol. 13, no. 5, pp. 1367–1376, 2013.
- [15] D. Broad, H. R. Maier, and G. C. Dandy, "Optimal operation of complex water distribution systems using metamodels," *Journal of Water Resources Planning and Management*, vol. 136, no. 4, pp. 433–443, 2010.
- [16] S. S. Hashemi, M. Tabesh, and B. Ataekia, "Ant-colony optimization of pumping schedule to minimize the energy cost using variable-speed pumps in water distribution networks," *Urban Water Journal*, vol. 11, no. 5, pp. 335–347, 2014.
- [17] F. K. Odan, L. F. Ribeiro Reis, and Z. Kapelan, "Real-time multiobjective optimization of operation of water supply systems," *Journal of Water Resources Planning and Management*, vol. 141, no. 9, p. 04015011, 2015.
- [18] M. Bazrafshan, N. Gatsis, M. Giacomoni, and A. Taha, "A fixed-point iteration for steady-state analysis of water distribution networks," in *2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, pp. 880–884, IEEE, 2018.
- [19] M. K. Singh and V. Kekatos, "On the flow problem in water distribution networks: Uniqueness and solvers," *IEEE Transactions on Control of Network Systems*, vol. 8, no. 1, pp. 462–474, 2020.
- [20] Y. Wang, C. Ocampo-Martinez, and V. Puig, "Stochastic model predictive control based on gaussian processes applied to drinking water networks," *IET Control Theory & Applications*, vol. 10, no. 8, pp. 947–955, 2016.
- [21] Y. Wang, V. Puig, and G. Cembrano, "Non-linear economic model predictive control of water distribution networks," *Journal of Process Control*, vol. 56, pp. 23–34, 2017.
- [22] H. Mala-Jetmarova, N. Sultanova, and D. Savic, "Lost in optimisation of water distribution systems? a literature review of system operation," *Environmental modelling & software*, vol. 93, pp. 209–254, 2017.
- [23] W. Liu, Z. Wang, X. Liu, N. Zeng, Y. Liu, and F. E. Alsaadi, "A survey of deep neural network architectures and their applications," *Neurocomputing*, vol. 234, pp. 11–26, 2017.
- [24] G. Guo, S. Liu, Y. Wu, J. Li, R. Zhou, and X. Zhu, "Short-term water demand forecast based on deep learning method," *Journal of Water Resources Planning and Management*, vol. 144, no. 12, p. 04018076, 2018.
- [25] M. Gavanelli, M. Nonato, A. Peano, S. Alvisi, and M. Franchini, "Genetic algorithms for scheduling devices operation in a water distribution system in response to contamination events," in *Evolutionary Computation in Combinatorial Optimization: 12th European Conference, EvoCOP 2012, Málaga, Spain, April 11-13, 2012. Proceedings 12*, pp. 124–135, Springer, 2012.
- [26] B.-G. Kim, Y. Zhang, M. Van Der Schaar, and J.-W. Lee, "Dynamic pricing and energy consumption scheduling with reinforcement learning," *IEEE Transactions on smart grid*, vol. 7, no. 5, pp. 2187–2198, 2015.
- [27] M. H. Moghadam and S. M. Babamir, "Makespan reduction for dynamic workloads in cluster-based data grids using reinforcement-learning based scheduling," *Journal of computational science*, vol. 24, pp. 402–412, 2018.
- [28] A. Candelieri, R. Perego, and F. Archetti, "Intelligent pump scheduling optimization in water distribution networks," in *Learning and Intelligent Optimization: 12th International Conference, LION 12, Kalamata, Greece, June 10–15, 2018, Revised Selected Papers 12*, pp. 352–369, Springer, 2019.
- [29] E. Ertin, A. N. Dean, M. L. Moore, and K. L. Priddy, "Dynamic optimization for optimal control of water distribution systems," in *Applications and Science of Computational Intelligence IV*, vol. 4390, pp. 142–149, SPIE, 2001.
- [30] Matlab Help Center, "Water Distribution System Scheduling Using Reinforcement Learning." <https://www.mathworks.com/help/reinforcement-learning/ug/water-distribution-scheduling-system.html> (accessed on September, 03, 2022).